

Aman Sharma

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Research Summary

Cross-disciplinary researcher combining **agentic AI systems**, **computational immunology**, and **protein engineering** to develop precision cancer immunotherapies. Experienced in CAR-T/BiTE engineering, single-cell genomics, and large-scale neoantigen discovery—bridging computational modeling with experimental immunoen지니어ing.

Education

- **Indian Institute of Technology, Madras, India** 2020 - 2022
 - *Master of Technology in Bio-Process Engineering* CGPA: 7.89/10.0
 - **Thesis:** "Optimization of DHA Production from Microalgae using Metabolic Flux Analysis"
- **Vellore Institute of Technology, Vellore, TamilNadu, India** 2016 - 2020
 - *Bachelor of Technology in Biotechnology* CGPA: 8.69/10.0
 - **Final Year Project:** "Degradation of Lambda-cyhalothrin by Bacterial Strains"

Research Experience

- **Perlmutter Cancer Center, NYU Langone Health, New York, USA** Sept 2023 - Present
 - *Assistant Research Scientist*
 - Developing **ImmunoVerse-Chat**, a conversational agentic AI that uses LLM reasoning to trigger multi-omics immunogenomic workflows and provide interactive, population-aware antigen discovery and therapeutic target prioritization.
 - Engineering cross-HLA **CAR-T cell** therapies using structure-guided design and recombinant protein expression, and developing optimized lentiviral/MLV vector production systems for transduction.
 - Implementing scalable **HPC workflows** using Docker and Singularity containers for neoantigen discovery from over 1,000 tumor samples and developing MS data preprocessing pipelines and immunogenicity scoring; I contributed these workflows to the **Pan-Cancer Intracellular Antigen Atlas** for systematic antigen prioritization across cancer types.
 - Performed **protein structural analysis** using PyMOL, VMD, and AlphaFold to guide CAR design optimization and predict antigen-TCR interactions.
- **IIT Madras, India & Institut Pascal UMR CNR, France** 2020 - 2022
 - *Collaborative International Research Project*
 - **DHA Production from Microalgae:** Optimized *Cryptocodinium cohnii* cultivation using FSEOF modeling, achieving **2-fold increase** in theoretical yield.
 - Developed **metabolic flux analysis models** to predict optimal growth conditions for enhanced lipid accumulation.
- **Indian Institute of Technology, Madras, India** 2020 - 2022
 - *Graduate Research Project*
 - **Modeling and Control of Cold Storage Thermal Stability:** A thermal dynamic model was created using **Simulink** for precise temperature control in agricultural cold storage.
 - Implemented **PID control algorithms** for automated temperature regulation in cold storage systems.

Technical Expertise

- **Computational Biology:** Python, R, MATLAB, PyMOL, VMD, AlphaFold, HPC workflows (SLURM), HLA typing (OptiType), RNA-seq analysis (DESeq2), single-cell analysis, pan-cancer genomics
- **AI & Automation:** LLM integration (OpenAI ChatGPT, Google Gemini), automated scientific workflows, prompt engineering for biomedical applications
- **Containerization & DevOps:** Docker containerization, Singularity for HPC, workflow management (CWL), version control (Git), automated pipeline development
- **Molecular & Cell Biology:** Protein expression/purification (FPLC, AKTA), CAR-T development, viral vectors (lentiviral/MLV), Gibson cloning, flow cytometry, PCR, cell culture, transfection/transduction
- **Analytical Methods:** UV-Vis/fluorescence spectroscopy, HPLC/GC chromatography, mass spectrometry, metabolic flux analysis (FSEOF), enzyme kinetics
- **Software & Tools:** ImageJ, FlowJo, GraphPad Prism, ChemDraw, Benchling, SnapGene, BLAST, TCGA portal, GEO database, CNCB-NGDC, LLM APIs

Publications & Presentations

- Li, G., Guzman-Bringas, O.U., **Sharma, A.**, Harpaz, Z., Tardio, E., Natarajan, A., Liu, X., Taylor, D., Yamin, R., Dellacherie, M., Krogsgaard, M., Cruz, C.R.Y., Yarmarkovich, M. (2025). "A pan-cancer atlas of T cell targets". [Link](#)
- **Sharma, A.**, et al. (2026). "ImmunoVerse-Chat: A conversational agentic-AI engine for next-generation immunotherapeutic target discovery". AACR Annual Meeting 2026, San Diego, USA. (Abstract accepted for presentation.)
- Jeelani, P., Ghai, A., Saikia, N., Kathed, M., Mitra, A., Krishnan, A., **Sharma, A.**, Chidambaram, R. (2020). "Baby Foods Based on Cereals" In: Gutiérrez, T. (eds) Food Science, Technology, and Nutrition for Babies and Children. Springer, Cham [Link](#)
- Batra, S., Adhikari, P., Ghai, A., **Sharma, A.**, Sarma, R., Suneetha, V. (2017). "Study and Design of Portable Antimicrobial Water Filter" *Asian J. Pharm. Clin. Res.*, 10(9), 268-271. [Link](#)

Awards & Honors

- Awarded **IIT-M x Polytech Clermont-Ferrand International Scholarship** for a collaborative research project optimizing microalgae growth and lipid extraction for enhanced DHA production (2021)
- Achieved **All India Rank 71** (Top 0.6%) in **GATE**(*Graduate Aptitude Test in Engineering*)- *Biotechnology* (2020)
- **Second Runner-up** at Siemens Healthineers Health-a-thon, VIT-Vellore for developing an innovative health-care solution addressing operational hazards in operating theaters (2019)

Teaching Experience

- **FIITJEE Ltd., Rajasthan, India** Dec 2022 - Jul 2023
 - Graduate Teaching Fellow - Biology
 - Mentored **over 80 medical aspirants** through personalized instruction, improving NEET/AIIMS exam performance by an average of 15%.
 - Designed and delivered a comprehensive biology curriculum covering molecular biology, genetics, and human physiology.
 - Conducted regular assessments and provided detailed feedback to track student progress.